

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

SEPTEMBER EXAMINATION

UCCD2303 DATABASE TECHNOLOGY /
UCCD2203 DATABASE SYSTEMS

WEDNESDAY, 4 OCTOBER 2023

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS

Instruction to Candidates:

This question paper consists of **Section A** and **Section B**.

Section A consists of **THREE (3)** questions. Candidates are required to **answer ALL** the questions.

Section B consists of **TWO (2)** questions. Candidates are required to **answer only ONE (1)** question. Should a candidate answer more than **ONE (1)** question in **Section B**, marks will only be awarded for the **FIRST** question in the order the candidate submits the answers.

Candidates are not allowed to use a calculator.

Answer questions only in the answer booklet provided.

UCCD2303 DATABASE TECHNOLOGY / UCCD2203 DATABASE SYSTEMS**Section A: Answer ALL QUESTIONS. [Total: 75 Marks]**

- Q1. (a) A foreign key references a primary key in the same table or another table. Predict what will happen when a foreign key containing null values and cannot match the values of a primary key in another table. (4 marks)
- (b) The multi-users access and sharing of resources by multiple interactive users at the same time may cause some problems. Explain **ANY THREE (3)** possible problems in a multiuser database system when the shared data is changed in the context of data insertion, deletion, and modification of relational database. (3 marks)
- (c) Consider the following scenario:
"A COURSE may be offered in the multiple CLASS SECTIONS."

Decide whether AGGREGATION or COMPOSITION exists in the given scenario above. Justify your answer(s). (6 marks)
- (d) Justify whether ONLINE ANALYTICAL PROCESSING is more efficient compared to ONLINE TRANSACTION PROCESSING. (6 marks)
- (e) In a database management system, data can be retrieved based on the requirements and can be easily filtered from the database. Argue **THREE (3)** reasons why some organisations would prefer using a traditional file-based system. (6 marks)
- [Total : 25 marks]

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Q2. (a) Consider the following relation R:

R (supplier_id, supplier_name, street, city, postcode, product_id, product_name, quantity, delivery_date_time)

Assuming that the relation R is used to store the product quantities and delivery date and time of shipments of products from suppliers. One supplier may supply the same product many times, each with a different delivery date and time.

- (i) Explain the term FUNCTIONAL DEPENDENCIES. (3 marks)
- (ii) Write the main FUNCTIONAL DEPENDENCIES (primary key dependency, partial dependency, transitive dependency and candidate dependency) for relation R. (5 marks)
- (iii) Formulate the First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF) for data in relation R. (8 marks)

(b) Consider the following relational schema for an insurance company in Figure 1 below:

OWNER(driver_id, license)
 CUSTOMER(driver_id, name, street, city)
 CAR(license, make, model, year)
 CLAIM(driver_id, license, claim_date, claim_description, claim_amount)

Note:

"City" is the attribute which contains information about a city e.g. "Los Angeles", "New York", "Kampar" etc.

"Make" is the attribute which contains information about the brand of the car e.g. Proton, Toyota, Honda etc.

Figure 1: A relational schema for an insurance company

- (i) Write **ONE (1)** relational algebra to display the names of all customers staying in "Los Angeles" or "New York" city. (4 marks)
- (ii) Write **ONE (1)** relational algebra to display all names of customers, street, and city of those car owners who bought "Toyota" car. (5 marks)

[Total : 25 marks]

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Q3. Answer the following questions based on the given scenario in Figure 2 below:

PetsXYZCare is a franchise company which provides healthcare services for domestic pets through various clinics located in the main cities of Malaysia. A clinic has many staff and one of the staff manages at most one clinic, not all staff manage clinics. The management would like to keep track of the history of all manager for each clinics. Each clinic has a unique clinic number. When a pet owner contacts a clinic, the owner's pet is registered with the clinic. An owner can own one or more pets, but a pet can only register with one clinic. The pet number is unique to a particular clinic. Some of the important details of the pet such as its current status (alive/deceased), and the details of the pet owner will be recorded. The clinics has a list of healthcare examinations for example, general checkup, flea infestation, or allergy healthcare examination. Owner can request for one or more healthcare examinations during the visitation.

When the pet comes along to the clinic, it undergoes the healthcare examination by a member of the consulting staff. After healthcare examination, the consulting staff will determine the treatment(s) needed. Based on the results of the healthcare examination of a sick pet, the vet may propose one or more types of treatment. The pet will be prescribed with one or more treatments after healthcare examination. The details of each treatment include a treatment number, a full description of the treatment, and the cost to the pet owner. The result and the date of the healthcare examination and the treatment will be recorded. Example of treatments are allergy injection, flu shot, flea spray, and etc. Charges for healthcare examination and treatment will be changed from time to time.

Figure 2: PetsXYZCare company scenario

- (a) Propose all the required primary key(s) and foreign key(s) for each of entity identified in Figure 2. (8 marks)
- (b) Construct **ONE (1)** Entity Relationship Diagram (ERD) from the given scenario above that include: all the entities, attributes of each entity, relationships, and the multiplicity of the relationship, i.e. cardinality and optionality. Hints: use CROW'S FOOT notation to illustrate your ERD. (8 marks)
- (c) Choose **ONE (1)** superclass/subclass relationship between any two entity types in the given scenario, whereby one entity type represents the 'whole' and the other is the 'part'. Justify your answer. (5 marks)
- (d) Determine whether SPECIALIZATION entities exists in the given scenario. Justify your answers. (4 marks)

[Total : 25 marks]

UCCD2303 DATABASE TECHNOLOGY / UCCD2203 DATABASE SYSTEMS**Section B Answer ONLY ONE (1) question. [Total: 25 marks]**

Q4. (a) Answer the following questions based on the given scenario in Figure 3 below:

ABC Bread Sdn. Bhd. is considering to migrate their existing database system into a distributed environment by using a Distributed Database Management Systems (DDBMS). Fragmentation is a database server feature to enable the control of the location of the data at the table level. As an IT consultant, you are required to provide some advice to the company before the implementation.

Figure 3: ABC Bread Sdn. Bhd.

- (i) Predict **ANY THREE (3)** possible concurrency control issues which could happen in the multi-user environment of ABC Bread Sdn. Bhd. (3 marks)
 - (ii) Justify **ANY TWO (2)** weaknesses of having fragmentation in distributed database. (4 marks)
 - (iii) The contents of a distributed database are spread across multiple physical sites. The contents may be stored in different systems that are located at one physical site or geographically far away. Suggest **ANY FOUR (4)** major DDBMS components which suit the needs of ABC Bread Sdn. Bhd. (8 marks)
- (b)
- (i) A spatial database is a database that is optimized for storing and querying data that represents objects defined in a geometric space. Discuss **ANY FIVE (5)** features how a spatial database is different from the common database systems. (5 marks)
 - (ii) Suggest **ANY FIVE (5)** real-life applications of spatial database. (5 marks)

[Total : 25 marks]

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- Q5. (a) Apache Hadoop is an open-source software framework that provides highly reliable for distributed storage and distributed processing of large data sets using simple programming models. Discuss how Hadoop can offer a flexible data management solution to meet Big Data challenges in a distributed computing environment. (6 marks)
- (b) Relate **SIX (6)** NoSQL database properties to performance and scalability issues in Big Data. (6 marks)
- (c) ACID characteristics can be categorized into four properties: atomicity, consistency, isolation, and durability. Depending on the need for robust consistency or availability, database designer may opt for either ACID-compliant databases or non-ACID-compliant databases. Explain how ACID characteristics affect database consistency and availability. (8 marks)
- (d) NoSQL databases are getting popular towards managing data for huge sets of distributed data. Argue **ANY FIVE (5)** reasons why database designer might not consider NoSQL database for implementation. (5 marks)
- [Total : 25 mark]
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